

Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-DLT-001: Load Customer and Order Data from CSV to Delta Tables

Related Functional Requirement(s): FR-DLT-001

Objective: Implement a Delta Live Table pipeline to read customer and order data from CSV files, perform data cleansing and transformation, and load it into delta tables.

Target Cluster Configuration:

- Cluster Name: Gen AI POC Databricks COE Cluster
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake, spark-csv

Source Data Details:

- Dataset Name: customerdata
- Location (Path/Table): `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata`
- Format: CSV
- Description: Customer data in CSV format
- Dataset Name: orderdata
- Location (Path/Table): `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`
- Format: CSV
- Description: Order data in CSV format

Target Data Details:

- Output Name: customer
- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.customer`
- Format: Delta
- Description: Customer data in delta format
- Output Name: order

- Location: `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.order`
- Format: Delta
- Description: Order data in delta format

Job Flow / Pipeline Stages:

- Read customer data from CSV file into a delta table `customer`
- Read order data from CSV file into a delta table `order`
- Add a new column `TotalAmount` to the `order` table
- Remove null and duplicate records from both `customer` and `order` tables

Data Transformations / Business Logic:

- Step: Read customer data from CSV file
- Description: Read customer data from CSV file into a delta table `customer`
- Transformation Logic: Use Spark CSV reader to read CSV file and write to delta table
- Step: Read order data from CSV file
- Description: Read order data from CSV file into a delta table `order`
- Transformation Logic: Use Spark CSV reader to read CSV file and write to delta table
- Step: Add new column `TotalAmount` to `order` table
- Description: Calculate `TotalAmount` as the product of `PricePerUnit` and `Qty`
- Transformation Logic: Use Spark SQL to add new column `TotalAmount` to `order` table

Error Handling and Logging:

- Log errors and exceptions to a log file
- Use try-except blocks to handle errors during data processing

TR-DLT-002: Create and Load ordersummary Table

Related Functional Requirement(s): FR-DLT-002

Objective: Implement a Delta Live Table pipeline to create an `ordersummary` table and load it with joined data from `customer` and `order` tables using SCD type 2 logic.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: customer
- Location (Path/Table): `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.customer`
- Format: Delta
- Description: Customer data in delta format
- Dataset Name: order
- Location (Path/Table): `gen\_ai\_poc\_databrickscoe.sdlc\_wizard.order`

- Format: Delta
- Description: Order data in delta format

Target Data Details:

- Output Name: ordersummary
- Location: `gen\_ai\_poc\_databrickscoe.sdmc\_wizard.ordersummary`
- Format: Delta
- Description: Joined data from `customer` and `order` tables in delta format

Job Flow / Pipeline Stages:

- Create `ordersummary` table if it does not exist
- Join `customer` and `order` tables on `CustId` and load the joined data into `ordersummary` table using SCD type 2 logic

Data Transformations / Business Logic:

- Step: Join `customer` and `order` tables
- Description: Join `customer` and `order` tables on `CustId`
- Transformation Logic: Use Spark SQL to join `customer` and `order` tables
- Step: Apply SCD type 2 logic
- Description: Update `StartDate` and `EndDate` columns accordingly to maintain history
- Transformation Logic: Use Spark SQL to apply SCD type 2 logic

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-003: Update ordersummary Table on Customer Data Change

Related Functional Requirement(s): FR-DLT-003

Objective: Implement a Delta Live Table pipeline to update the `ordersummary` table whenever there is a change in the `customer` table, using SCD type 2 logic.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: customer
- Location (Path/Table): `gen\_ai\_poc\_databrickscoe.sdmc\_wizard.customer`
- Format: Delta
- Description: Customer data in delta format

Target Data Details:

- Output Name: ordersummary
- Location: `gen\_ai\_poc\_databrickscoe.sdmc\_wizard.ordersummary`
- Format: Delta

- Description: Updated `ordersummary` table in delta format

Job Flow / Pipeline Stages:

- Identify changes in the `customer` table
- Update the `ordersummary` table by making old records inactive and new records active

Data Transformations / Business Logic:

- Step: Identify changes in `customer` table
- Description: Identify changes in `customer` table
- Transformation Logic: Use Spark SQL to identify changes in `customer` table
- Step: Update `ordersummary` table
- Description: Update `ordersummary` table by making old records inactive and new records active
- Transformation Logic: Use Spark SQL to update `ordersummary` table

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-004: Create and Load customeraggregatespend Table

Related Functional Requirement(s): FR-DLT-004

Objective: Implement a Delta Live Table pipeline to create a `customeraggregatespend` table and load it with aggregated data from the `ordersummary` table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: ordersummary
- Location (Path/Table): `gen\_ai\_poc\_databrickscoe.sdsc\_wizard.ordersummary`
- Format: Delta
- Description: Joined data from `customer` and `order` tables in delta format

Target Data Details:

- Output Name: customeraggregatespend
- Location: `gen\_ai\_poc\_databrickscoe.sdsc\_wizard.customeraggregatespend`
- Format: Delta
- Description: Aggregated data from `ordersummary` table in delta format

Job Flow / Pipeline Stages:

- Create `customeraggregatespend` table if it does not exist
- Aggregate `TotalAmount` column from `ordersummary` table by `Name` and `Date`

Data Transformations / Business Logic:

- Step: Aggregate `TotalAmount` column
- Description: Aggregate `TotalAmount` column from `ordersummary` table by `Name` and `Date`

- Transformation Logic: Use Spark SQL to aggregate `TotalAmount` column

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-005: Implement using Delta Live Tables

Related Functional Requirement(s): FR-DLT-005

Objective: Implement the above functionality using Delta Live Tables.

Target Cluster Configuration: Same as TR-DLT-001

Job Flow / Pipeline Stages:

- Implement the data pipelines using Delta Live Tables to perform the required data transformations and loading

Data Transformations / Business Logic: Same as TR-DLT-001 to TR-DLT-004

Error Handling and Logging: Same as TR-DLT-001